

Should I stay or should I go?
Living standards of serfs and peasants in 18th century Moscow.

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Abstract: Russian serfdom is often compared to slavery, as both systems kept the majority of the population at subsistence levels (Acemoglu & Robinson, 2012). However, empirical evidence supporting this comparison is limited. Robert Allen's micro-level methodology, which compares living standards by analyzing wages and prices, provides a new approach to understanding the economic conditions of the poorest groups. Recent studies of slavery show a significant decline in living standards for slaves, placing them below subsistence levels (Burnard et al., 2017). Yet, this methodology has not been applied to Russian serfs during their peak period. This study fills this gap by examining whether it was more advantageous for serfs to farm semi-arable land or work as laborers in pre-industrial Moscow. Using archival data from 1715 to 1815, I analyze seasonal wages and purchasing power. The findings show that urban wages were lower than agricultural incomes. Unlike slaves, serfs had the autonomy to make economic choices, which affected both the Moscow economy and peasant households. The volatility in wages and prices demonstrates the rational economic motivations behind serfs' decisions. This study highlights the importance of considering household life cycles in non-industrialized societies and shows how regional differences, such as those seen in Moscow during the Napoleonic Wars, are crucial for understanding broader economic trends.

Key words: Russian Serfdom, Slavery, Living Standards, Pre-Industrial Russia, Regional economic differences.

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Russian serfdom is often compared to slavery; both extractive institutions kept most of the population at a subsistence level (Acemoglu and Robinson 2012). However, this common belief lacks empirical evidence. Only recently Robert Allen's methodology made it possible to compare different economies across various periods from a micro-level perspective, which is essential for understanding the living standards of the poorest social groups. Recent research on slavery has confirmed the significant negative impact on living standards, with slaves' well-being falling significantly below subsistence level (Burnard et al., 2017). However, this methodology has not yet been applied to Russia during the heyday of serfdom. More quantitative micro-level research using a standard methodology is needed to understand how serfs lived under serfdom and how it differed from slavery concerning living standards. I propose addressing this broad issue by asking a very specific question: was it more profitable for a serf to farm semi-arable land or work as an unskilled or semi-skilled labourer in a city, such as Moscow, before industrialization?

I measure the living standards of seasonal workers (*otkhodniki*) in Moscow and its surroundings in the long 18th century (1715 to 1815) by extracting salary and price data from previously unstudied archival materials. Using Allen's approach, I compare the purchasing power of workers' wages over time to assess their living standards. I calculate the ratio of annual incomes of unskilled laborers to the cost of a subsistence basket, adjusted for national consumption patterns. The results align closely with the original thoughts of contemporaries and the Russian government, confirming the validity of Allen's subsistence basket concept. This corroboration allows us to meaningfully understand the role seasonal work earnings played in the peasant household economy. The findings reveal a negative urban wage premium, showing that annual income from unskilled work in Moscow was lower than that from agriculture in surrounding areas.

I argue that, unlike slaves, serfs could make their own decisions and move to the city to do so they needed to buy a passport. Their mobility significantly impacted both the economy of the capital and the peasant household economy. In 1785, temporary workers made up 40 percent of the workforce (RGADA, f. 1261, op. 12, d. 100). The paper highlights the dynamism of the serfdom economy in Moscow and the active agency of serfs and peasants. The volatility of wages and prices reflects the economically motivated behaviour of serfs and peasants. They could either maintain a subsistence level by performing agriculture or move to the city and engage in non-agricultural activities. Sending someone to the city was rational only when there was an extra person in the household, such as an unmarried son. This research highlights the importance of understanding the household life cycle in studies of living standards in non-industrialized societies. By focusing on a specific area, I gain a better understanding of its context. Interestingly, the latest GDP per capita series by Broadberry and Korchmina show stagnation from the late 18th century to the 1870s, without any obvious impact from events like the Napoleonic Wars. However, Moscow was significantly affected by these wars, as seen in changes in the welfare ratio of laborers, which show the regional impact of this nationally important event. This underscores the importance of regional perspectives, as overall GDP data for Russia does not reflect these differences.

Moscow Province was chosen for two reasons: first, due to the availability of data, and second, because it was located at the center of Russia's serfdom lands. Moscow, the 'Enlightened Metropolis' and one of the capitals of the Russian Empire, underwent significant transformation in the long 18th century (Martin 2013). Construction booms created a substantial job market that attracted freelance seasonal workers, including semi-skilled workers (such as carpenters and joiners) and unskilled workers (such as diggers and porters). These industries were the first to respond to economic conditions, and Moscow, as the largest Russian city, saw the most intense economic activity. Therefore, Moscow serves as an excellent example of the potential opportunities available to serfs outside of agriculture. If there were any potentially positive aspects of serfdom in Russia, they could likely be observed in Moscow.

I apply the approach developed by economic historian Robert Allen. I calculate the ratio of annual incomes of unskilled laborers to the costs of a subsistence basket adjusted to national consumption patterns. In the existing literature, laborers are traditionally the main research subjects in comparative studies on wages. First, the data on their daily earnings provide a clear spatial and temporal perspective for comparisons. Second, the level of prosperity available to unskilled workers represents the very basic standard of living achieved in the society under study, providing a clear understanding of how favourable

the economic environment was. This approach is valuable because it allows us to assess living standards even in a data-sparse environment.

We use data from ledgers of construction project expenditures from private and public institutions in Moscow. We focused on financial documents from large monasteries and private estates (e.g., Znamensky Monastery, Ugresh Monastery, Trinity Lavra of St. Sergius, Golitsyn estates). These institutions and landowners consistently invested in building and refurbishing their properties in Moscow. Due to the scale of their economies, they regularly purchased products, and their revenue and expenditure books systematically recorded prices and wages. These records reflect actual purchasing patterns, such as the absence of rye prices in June and July but the presence of beef and butter prices during the harvest in July and early August when mowers were hired and provided with food. We also include price data from Moscow newspapers when local authorities began publishing them monthly. However, despite the abundance of records, data on daily wages and certain important goods (e.g., cheap cloth, soap, firewood) are sparse. Most wages were paid per task or per unstandardized unit rather than per day. Consequently, most records cannot be used for wage series, explaining why we have a complete set of prices for all goods in the subsistence basket and daily wages for only 29 out of 100 years.

The results show that the economic development of the Moscow region experienced both growth and decline. While nominal wages and prices were rising, the welfare ratio of seasonal workers was highly volatile. It skyrocketed during construction booms, reaching an unprecedented 5.1 after the fire of 1812, but this growth never lasted more than a decade. In contrast, annual agricultural income in surrounding areas provided a stable subsistence level, unlike the fluctuating income from unskilled work in Moscow. Therefore, it was more rational for serfs and peasants to spend most of their time cultivating even semi-suitable land rather than seeking quick money in the city. This pattern of “quick money” underscores the unsustainable nature of the region's economic development.

The paper contributes to several sets of literature.

First, I incorporate Russian data into comparative research on wages and prices, a topic that gained renewed interest in the early 21st century. There was an emergence of publications aiming for comparison of the purchasing power of the wages in different regions for long scale time periods. A significant milestone, which was Robert Allen's article, presented calculations that showcased how the gap between the ‘real wages’ in Great Britain and the Netherlands, on one hand, and the rest of Europe on the other, emerged in 1500 – 1750 (Allen 2001). Calculations based on similar methods were carried out for Italy, North American colonies of Great Britain, New France (modern Quebec), Denmark, Poland, China, etc (Malanima 2013, Allen et al 2012, Geloso 2015, Khaustova and Sharp 2015, Malinowski 2016, Allen 2011). The composition of the baskets is adjusted according to the country's characteristics. The changes primarily targeted the main crop or substances. Naturally, the researchers do not claim that made-up baskets represent people's consumption behaviour precisely. But this research creates a necessary benchmark for international comparison.

Second, this paper brings us to better understanding of the economic state of Russian peasantry in the 18th century, particularly in terms of incomes and consumption patterns. Despite historians' efforts, we currently have only scattered series of prices or wages for short periods and for very specific types of skilled workers (Mironov 1985; Strumilin 1954; Voronov 1955; Pavlenko 1962; Zaozerskaia 1949; Kovalchuk 1999; Hellie 1999; Nikolaeva 2004; Shipilov 2008; Voronov 1955; Kovalchuk 1999; Dennison and Nafziger 2013; Smith 2008). The only index of Russian real wages for the 18th century, corresponding to purchasing power parity but not conventional methodology, was created by Boris Mironov (Mironov 2010). Given the differences in methods and data, it is not surprising that researchers' findings on wage dynamics in the 18th century vary significantly and are not always replicable due to the lack of original data publication, often leaving us with aggregated or otherwise manipulated data. A recent paper by Robert Allen and Ekaterina Khaustova includes calculations that can be directly compared to ours due to the similarity in methods (Allen and Khaustova, 2019). This highlights the existing gaps in data and meaningful conclusions about the economic development of 18th-century Russia. Our contribution lies in both data publication and advancing the understanding of the economic realities of that period.

Beyond Bondage: The Role of Serf Mobility and Economic Dynamics in Russian Peasant Society.

Serfdom concentrated political power within a small landed elite while imposing few constraints on their authority, which had significant implications for economic institutions. Many authors argue that, due to limited control over their labour, serfs faced distorted incentives that hindered the efficient allocation of resources and ultimately slowed economic growth (e.g., North and Thomas 1973; Acemoglu and Robinson 2012).

In contrast, some scholars—especially those studying the relationships between landlords and serfs—highlight the dynamic nature of serfdom as an institution that could either promote economic development or prevent economic decline (e.g., Moon 1996; Dennison 2011; Stanziani 2014; Malinowski 2016). They argue that landlords played a crucial role in maintaining social order and providing serfs with minimal food security, particularly during times of famine. Moreover, despite various restrictions on their mobility, industrious serfs could still find opportunities for economic advancement within the system.

It is important to remember that the peasantry was not a uniform social class in Russia. It varied over time in terms of the social groups within it and the proportions of different groups. The most well-known group in Russian peasantry is the serfs. Nancy Kollman provides a precise characterization of Russian peasantry: serfdom did not define the Russian Empire, as serfs made up about half of the peasant population and were primarily concentrated in central provinces, including Moscow (Kollman 2017). In other areas, peasants lived without the oversight of a landlord and were directly subordinate to the crown, the Church, or the College of Economy (for former Church peasants after 1764). These peasants, often called state peasants or “unserfed peasants,” were mostly found in the empire’s borderlands: the north, the northern Urals, Siberia, and the southern steppes, where various military service groups had merged with the peasantry. Peasants personally owned by the imperial family declined from 9 percent to 5 percent of the peasant population over the eighteenth century and were scattered across the empire, as were the thousands of peasants transferred to state ownership from the Church in the 1760s (known as Treasury or Economic peasants). By the end of the eighteenth century, state peasants made up 20 percent of the peasantry in central regions but nearly half in the northern and southern borderlands.

One of the key issues concerning Russian serfdom is the mobility of serfs. While the conventional view holds that serfs were tied to the estates and worked solely on their landlord’s land, historical evidence suggests that Russian serfs exhibited notable mobility and were even permitted to temporarily relocate, including to Moscow. By the mid-18th century, the percentage of seasonal laborers in the Central Non-Black Earth region was already considerable. According to M.F. Prokhorov, in eight districts of what would later become Moscow province (Borovsk, Volokolamsk, Zvenigorod, Klin, Kolomna, Moscow, Ruza, and Serpukhov), approximately 6% of male peasants worked as seasonal laborers (Prokhorov 1997, Table 35). By the century’s end, this figure had increased, with estimates suggesting that between 10% and 15% of the male peasant population in Moscow province were engaged in seasonal work, while in Yaroslavl province, the percentage ranged from 15% to 23% (Chernikov 2013, p. 224). This phenomenon can be interpreted in two ways: either serfs and peasants were motivated to seek work in Moscow on their own, or landlords were sending them there. Since there is no systematic evidence that landlords were intentionally trying to offload their serfs, it’s likely that serfs were choosing to migrate voluntarily.

These estimates are based on data from issued passports, which were required for peasants travelling more than 35 kilometres (30 versts) from their place of residence. A key indicator of the growing significance of seasonal labour is the April 1758 law that extended passport validity for European peasants travelling to Siberia (PSZ RI-I, № 10829). This suggests that populations in various Russian regions were more mobile than is typically assumed in stylised papers when discussing serfdom. Moreover, by the 18th century, “including unfree peasants from towns and villages and bonded serfs, over three-quarters of the population in the industrial centre of the Moscow region could sell their labour to any employer without legal barriers” (Strumilin 1954, p. 264). However, these estimates may be somewhat optimistic. I still agree with V. A. Fedorov, who noted that serfs constituted the majority of Moscow’s seasonal labourers (Fedorov 1997, p. 174).

It is important to view the passport system not as a restrictive measure, but rather as a way of tracking taxpayers. This perspective helps explain why the costs associated with obtaining a passport were relatively low. The cost of a passport for up to one year increased gradually: from 0.02 rubles in the 1720s to 0.1 rubles in the 1760s, then to 1.0 ruble in the 1790s, and by the 1830s, it reached 1.45 rubles (PSZ-RI-I, Vol. 7, No. 4533; Vol. 8, No. 5226; Vol. 16, No. 11988; Vol. 23, No. 17226; Svod 1857, p. 39). When compared to daily wages, the affordability of a passport is even clearer. In the 1720s and 1760s, the cost of a passport was equivalent to about one day's wage in Moscow, and by the 1790s, it had risen to the cost of about four days' wages.

Furthermore, the situation regarding the relatively high level of mobility among serfs and peasants remained consistent across both the 18th and 19th centuries and throughout various provinces. As Smith (2008) indicates for Kostroma province in 1840, out of a total provincial population of nearly 50,000 male serfs, officials issued 5,000 yearly passports (10%) and an additional 10,000 monthly passports (20%) for work away from their land.

Subsistence Basket Moscow in the long 18th century (1700 – 1812)

As mentioned above Allen's subsistence basket usually requires some adjustment to local products (Allen 2001). We chose rye as the main food product. This crop was undoubtedly the staple food in European Russia. Its prices have been studied to the greatest extent and are closely correlated with the prices of rye flour (Kovalchenko and Milov 1984). We exclude legumes from the basket and increase the amount of rye so that it could provide enough calories for survival. Of the products that are usually absent from the basket, salt is included. Salt was an essential product in climate like Russian. Heating fuel is calculated based on the firewood consumption rates prescribed for heating of barracks and cells, which are significantly higher than in Allen's subsistence basket for the UK (approximately 12.4 against 3 million BTU).¹ Our basket is calculated for 2,100 Cal (table 1²).

Table 1. Annual subsistence basket for a person for the Russian 18th century.

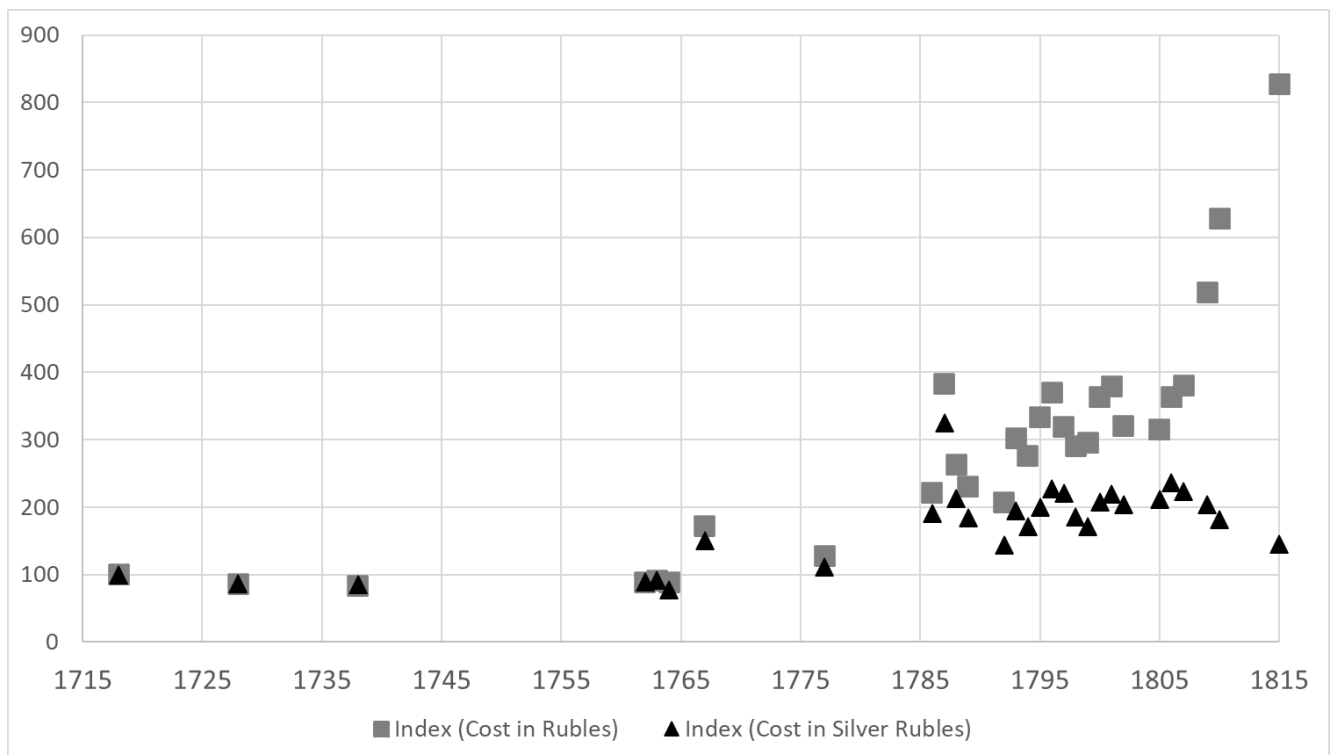
Product	Quantity per person per year (metric units)	Quantity per person per year (original units of the 18c.)
Rye	276.5 kg	2.09 chetverts
Beef	5.0 kg	12.19 pounds
Butter	5.0 kg	12.19 pounds
Salt	7.0 kg	0.42 poods
Soap	1.3 kg	0.08 poods
Linen cloth	5.0 m	7.04 arshins
Tallow candles	2.6 kg	0.16 poods
Firewood	2.5 m ³	0.26 sazhen
Rent	5%	5%

Figure 1 shows the results of the subsistence basket calculations. The data indicates that the nominal cost of the subsistence basket increased nearly tenfold over 100 years, from 4.77 rubles in 1718 to 39.39 rubles in 1815. However, when adjusted to silver prices, there is no significant increase, with costs rising only slightly from 0.98 in 1718 to 1.42 in 1815.

¹ Polnoe sobranie zakonov Rossiiskoi Imperii. Sobranie II [Complete Collection of the Laws of the Russian Empire (First Collection)] (hereafter - PSZ), vol. 33, part 1, no 33073. Similar norm of firewood consumption per head (2,4 m³) was reported for Moscow in 1921 (A.V. Dulov, *Geograficheskaya sreda i istoriya Rossii. Konets XV – seredina XIX veka* [Geographical Environment and Russian History From the End of the 15th to the mid-19th Century] (Novosibirsk, 1983), 134). I assumed that one m³ of firewood weighs 500 kg and that 1 kg of firewood gives 2500 kcal.

² Skurikhin, I. M., & Tutelyan, V. A. (Eds.). (2002). *Chemical composition of Russian food products: Reference guide*. DeLi Print. Verified by calculations based on Clark and Haswell (1967), upper boundary.

Figure 1. Indexed Cost of Subsistence Basket in Rubles and Silver Rubles (1718=100)³.



Sources: Appendix 1.

Between 1718 and the 1760s, we see that both indices remain close to the base value of 100, indicating stable subsistence costs in both nominal and silver rubles. From the 1760s to the 1780s, there is a moderate increase in the nominal ruble index, suggesting rising prices. Starting around 1780, there is a significant increase in both indices, especially in nominal rubles, which reach much higher values by 1815. This indicates rapid inflation in nominal terms, while the silver ruble index remains relatively stable. It is important to consider price growth in both nominal and real terms when assessing the decision-making of serfs. The doubling of prices at the end of the 18th century and the beginning of the 19th century likely had a major impact on serfs' strategies, influencing their choice between moving to the city for work or staying in the village to farm.

The data obtained can be verified by calculating the average cost of the subsistence basket using Mironov's data (Mironov and Lindert). The database he compiled contains the maximum, the minimum, and the average price for a product per decade. When calculating the subsistence basket based on these data, I used more accurate data on rye prices collected by the same Mironov (the average price being calculated based on the entire set of prices over a decade, not the middle value between the maximum and the minimum) (Mironov 1985, table 3). I also use our own data on the price of the cheapest cloth, since Mironov's database contains prices for a relatively expensive canvas. The resulting average decennial costs of subsistence baskets are not different. (See Table 2.)

³ Nominal prices are calculated based on transaction records from revenue and expenditure books, which were recorded only in kopeks due to the small values involved. For international comparisons, I converted these prices into grams of silver per ruble. This conversion adjusts for changes in the silver content of one silver ruble over time, using conversion rates documented by contemporaries.

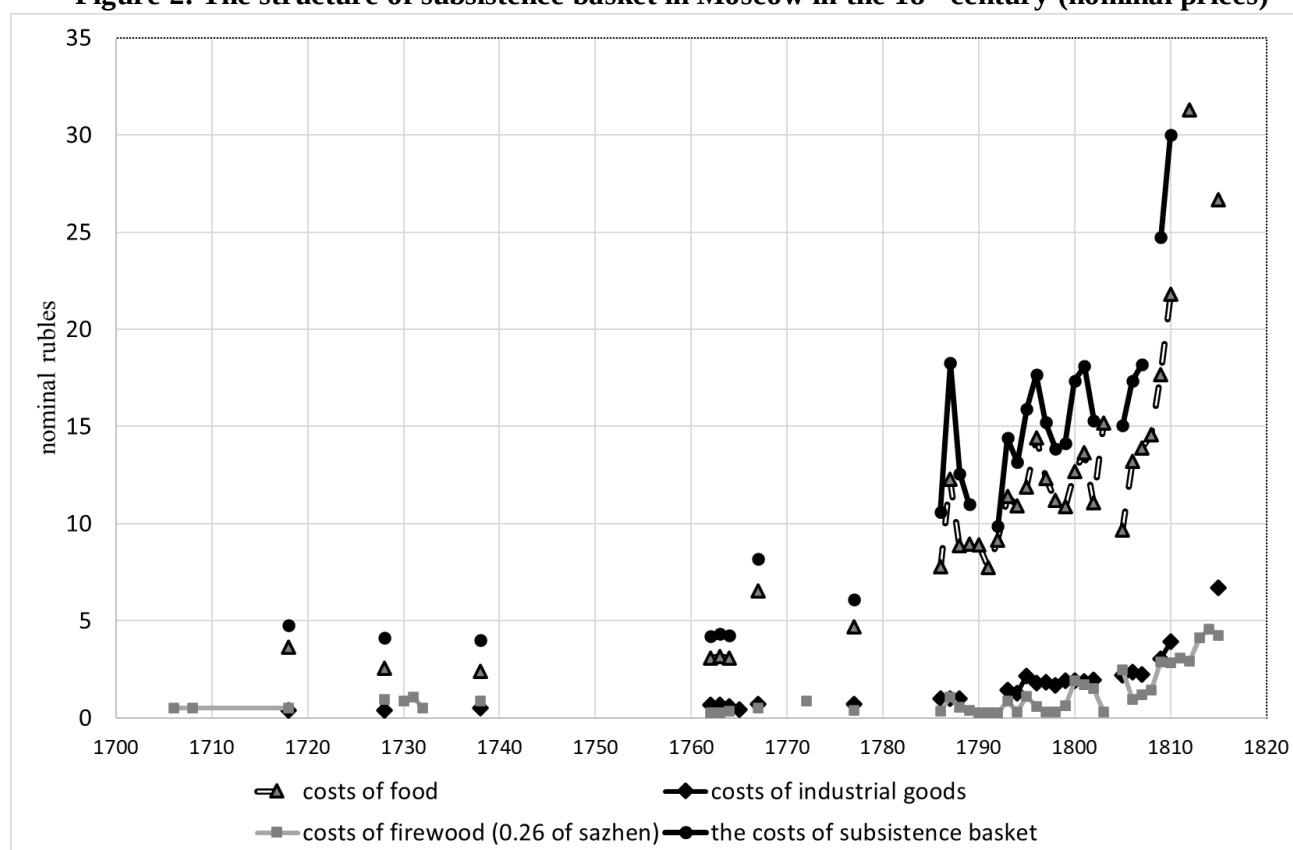
Table 2. Comparative Analysis of Average Basket Costs (1760s - 1790s).

	Average Cost (Mironov's Data) (rubles)	Average Cost (Our Data) (rubles)	Percentage Difference (%)
1760s	5.94	5.24	- 11.8
1770s	7.01	6.12	- 12.7
1780s	10.77	13.11	21.7
1790s	14.75	14.28	-3.2

Sources: Appendix 1; Boris Mironov, Peter Lindert, Russia. Prices and wages 1590 – 1871, *Global Price and Income History Group*: <http://gpih.ucdavis.edu/Data/filelist.htm>.

The table presents a comparative analysis of average basket costs from the 1760s to the 1790s, showing that while our data indicated lower costs than Mironov's in the earlier decades, the 1780s experienced a significant increase in our data, leading to higher costs compared to Mironov's, before slightly decreasing in the 1790s.

What caused the increase in basket costs over time? Figure 2 indicates that it was primarily due to the grocery portion of the basket.

Figure 2. The structure of subsistence basket in Moscow in the 18th century (nominal prices)

Sources: Appendix 1.

As we will demonstrate later, the quantity of products in the basic subsistence basket is lower than what people in the 18th century considered sufficient. However, the included goods were indeed important for Moscow residents.

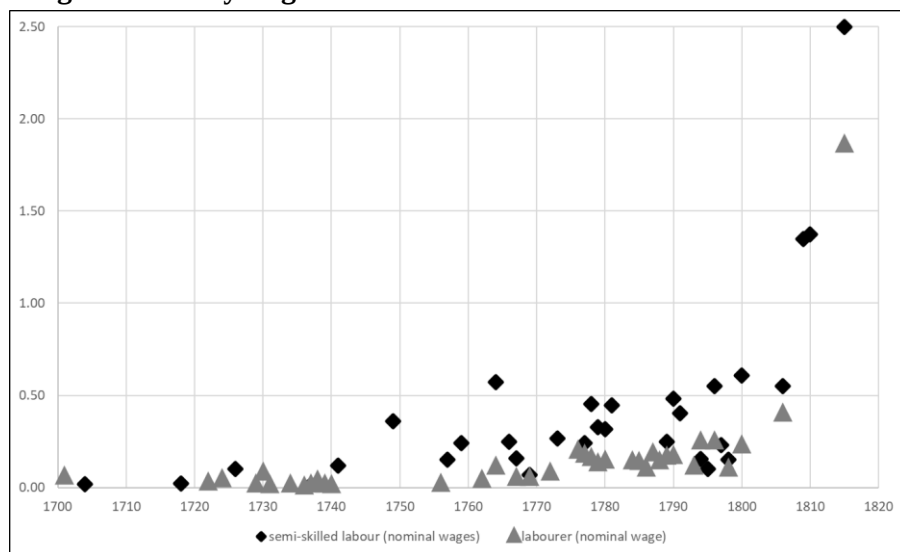
Wages of semi-skilled and unskilled construction workers

In this section, we discuss the daily earnings of unskilled workers who performed tasks such as digging, removing construction debris, breaking stones or bricks, and cobbling pavements. Workers hired in Moscow were mainly serfs and state peasants. The construction season in Moscow, as in most of Russia, had its peculiarities; the cold winter made construction either impossible or extremely

inefficient. This construction season coincided with key periods of agricultural work, preventing the involved peasants from adequately engaging in agriculture. Therefore, we classify all construction workers into semi-skilled (carpenters, bricklayers, brickmakers), as becoming a carpenter, for instance, requires time and specialisation. During the period under review, most peasants preferred to limit seasonal work to the period from autumn to spring, whereas in summer, there were fewer seasonal workers and labour wages were slightly higher (Fedorov 1997). In conclusion, semi-skilled and unskilled laborers were defined based on the nature and complexity of their tasks. Unskilled workers performed basic, physically demanding jobs. In contrast, semi-skilled laborers, such as carpenters, bricklayers, and brickmakers, required time and specialization to develop their skills.

Figure 3 illustrates the distribution of nominal and real (adjusted for silver) daily wages of semi-skilled and unskilled workers in Moscow during the 18th century. It reveals a significant range in the data.

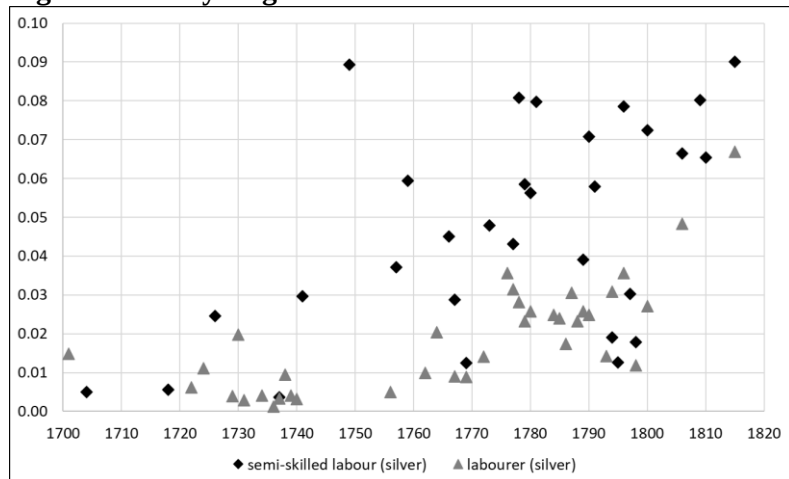
Figure 3a. Daily wages of labourers and construction workers in nominal rubles in Moscow in the 18th c.



Sources: see Appendix 2.

Figure 3a shows that in the mid-18th century, the nominal daily earnings of unskilled and semi-skilled construction workers rose rapidly. This growth can be explained by several factors. Firstly, the increasing importance of specialization made qualified specialists scarce, despite the growing number of seasonal workers. Secondly, a shortage of specialists during a period of heightened demand for construction drove wages up. Since the mid-18th century, the nobility and the state had been striving to bring their houses and cities up to European standards. This led to active private and public construction, especially after the great fire of 1773 (Voronov 1957). After the war of 1812, the wage increase was even more significant, with semi-skilled workers earning 1.30 roubles in 1810 and 2.5 roubles in 1815. Additionally, the difference in nominal wages of labourers and semi-skilled workers widened.

Figure 3b. Daily wages of labourers and construction workers in nominal rubles in Moscow in the 18th c.



However, there was no notable increase in wages when adjusted for silver (figure 3b). As mentioned earlier, it is crucial to consider changes in nominal values for decision-making of serfs and peasants.

Table 3 shows the indices of nominal daily wages in two capitals of Russian Empire in the 18th century. We are incorporating our data with the indices from Boris Mironov (Mironov 2010).

Table 3. Indices of nominal daily wages in Moscow and Saint Petersburg

	Daily wages of unskilled construction laborers in Moscow	Daily wages of semiskilled construction laborers in Moscow	Daily wages of construction low skilled workers in Saint Petersburg
1710s		1.19	25.40
1720s	1.54	5.16	33.95
1730s	1.49		36.72
1740s	0.70	12.39	26.10
1750s	1.08	10.06	30.48
1760s	3.41	14.06	31.64
1770s	8.01	16.64	35.10
1780s	7.76	17.43	51.04
1790s	9.48	15.81	67.90
1800s	18.82	43.14	92.61
1810s	100	100	100

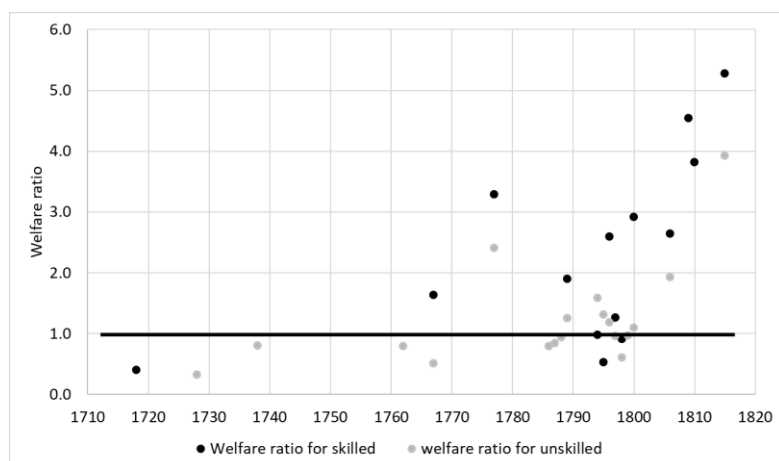
Sources: Mironov 2010, p. 56, Appendix 3.

Table 3 shows that the level of growth of ‘joiners’ wages in Saint Petersburg and in Moscow. However, starting from the mid-18th century, both unskilled and semi-skilled laborers in Moscow experienced a higher rate of nominal wage growth compared to joiners in Saint Petersburg. This can be attributed to the construction boom in Moscow after the 1770s, especially considering that Moscow was somewhat neglected as a metropolitan city in the first half of the 18th century. The substantial wage growth in the 1810s, compared to the previous decade, was driven by the need to rebuild Moscow after the fire of 1812. This reflects the symbolic importance of Russia's two capitals during the long 18th century. Methodologically, this highlights the limits of generalizing results from one city to the whole country, especially one as large as Russia.

Welfare ratios

We calculated the welfare ratio for Moscow. The solid line in Figure 4 represents the ratio between daily earnings for 250 working days per year and the costs of three annual basic baskets, which could sustain one breadwinner, his wife, and two children. Dots above the line indicate earnings sufficient to support additional dependents.

Figure 4. Welfare ratio for labourers in Moscow.



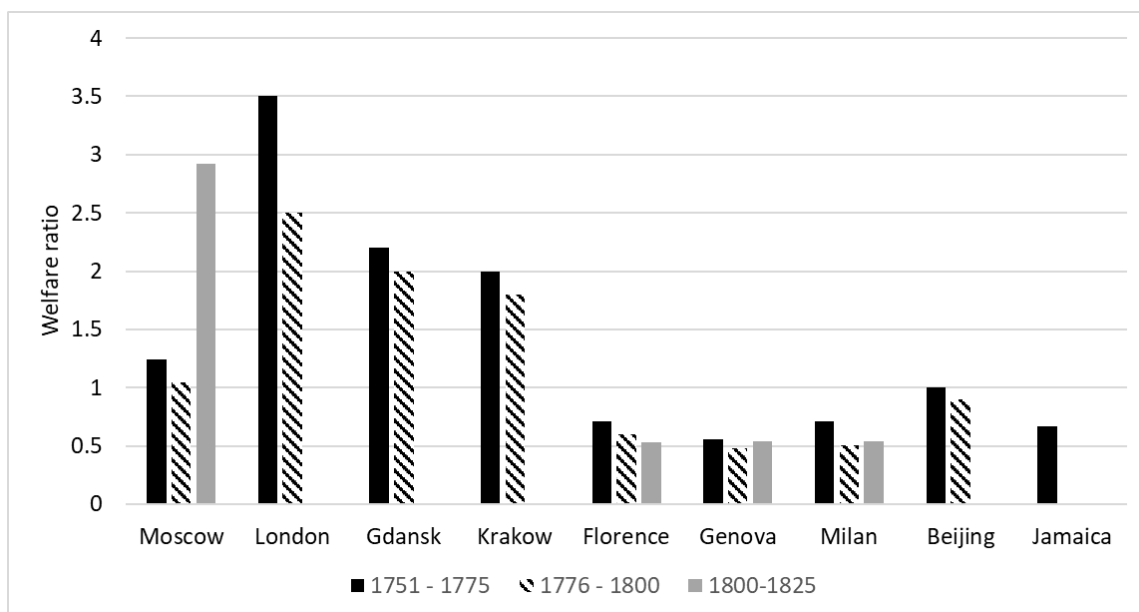
Source: Appendix 3.

The welfare ratio for unskilled laborers in Moscow shows significant volatility. The economic conditions were less favourable in the second half of the 1780s, when laborers earned daily wages insufficient to support a family. This illustrates that periods of welfare ratio growth were short-lived, as after a sharp increase, subsequent “bottoms” were slightly higher than the previous one.

How did Moscow compare to other cities? (Figure 5)

Figure 5 shows that in the second half of the 18th century, living standards in Russia were adequate for supporting a family, even though the country appeared economically poorer than the rest of Europe. The notable spike in the welfare ratio during the 1800s and 1810s is striking; however, it's important to recognize that this increase was temporary. The graph also includes Jamaica, which presents a welfare ratio that averages between slaves and poor whites for just one year, revealing that the welfare ratio for slaves was only 0.4. This comparison of living standards between slavery and serfdom clearly indicates that serfs were in a significantly better position than slaves.

Figure 5. Welfare ratio for unskilled workers.



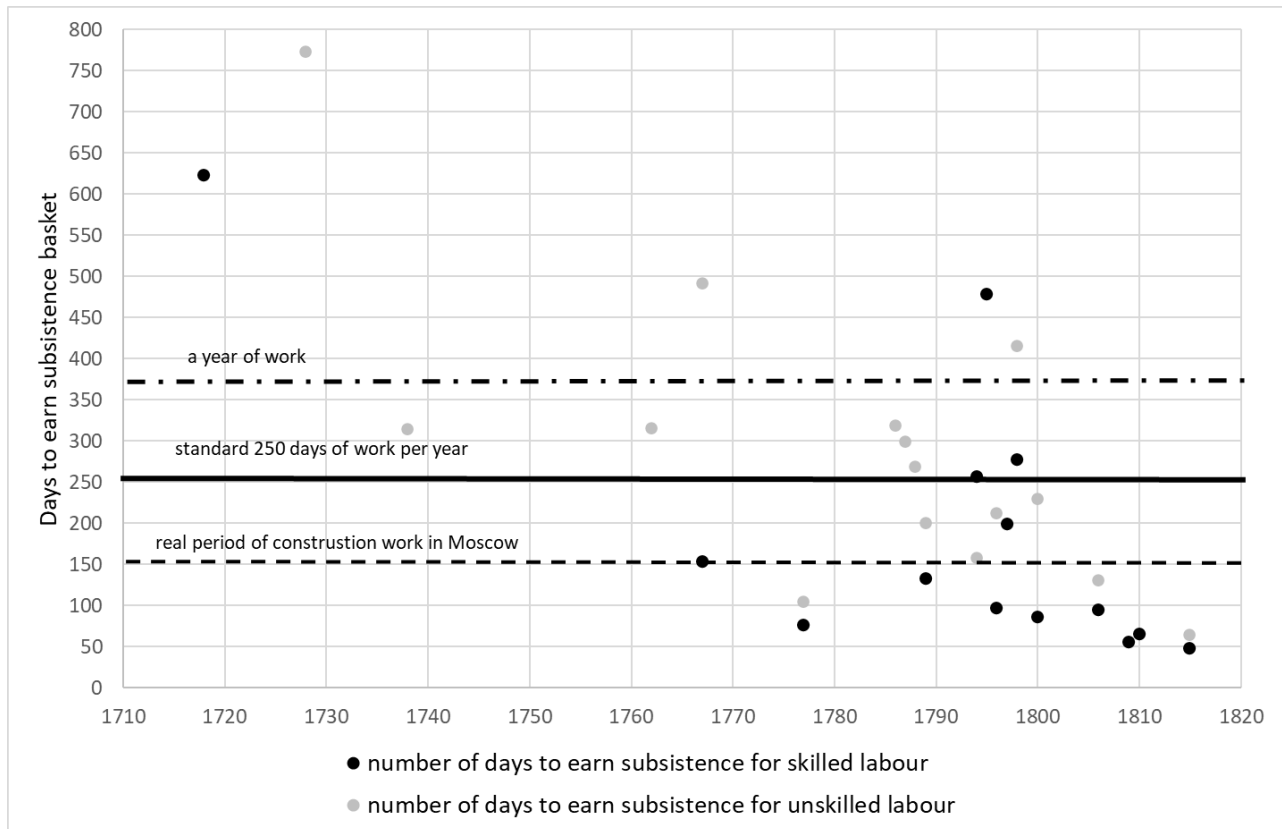
Sources: Allen 2009, p. 40; for Poland: Malinowski 2016, p. 7, Table 4; for Jamaica: Burnard et al., 2017, Table 2, Appendix 3. For Italy, we use averages for the years 1750–1775 and 1776–1800 (Cappelli et al., 2024).

However, the question remains whether it was feasible to work 250 days a year in Russia, as construction and agricultural periods overlapped significantly, leading serfs and peasants to work intensely for shorter periods. As Jane Humphries and Jacob Weisdorf (2019) show for England, this could significantly distort the results.

The contracts for construction work we found typically covered the period from May to October. Payments for construction work in the income and expense books were also recorded during this period. For instance, construction work at the Bolshoi Nikitsky house and the country houses on Devichye Pole, owned by A. M. Golitsyn, occurred from May 24 to October 27, totalling 151 days. The records also detailed days when no work occurred, such as Sundays and major church holidays like June 29 (Peter's Day) and July 8 (Day of the Kazan Icon of the Mother of God). Out of the 151 days, 20 were weekends and 10 were holidays, resulting in only 121 working days (RGADA, f. 1263, op. 1, d. 6444; 6224; f. 292 op 1 kn. 4, l. 563).

If we assume that construction work primarily took place from May 1 to October 31, excluding holidays and weekends, the season will span 184 days a year. In such a scenario, the actual number of working days could approximate around 150.

Figure 6. Number of Days Required to Earn a Subsistence Basket for Skilled and Unskilled Construction Workers in Moscow in the 18th Century.



Source: Appendix 1, 2 and 3.

Figure 6 shows that it became possible for unskilled labourers in Moscow to earn enough money in construction only at the beginning of the 19th century. For skilled workers, this became more achievable after the 1770s when Moscow garnered sufficient attention from the central government, and the nobility began living there permanently, taking advantage of the flexible state service initiated during the time of Catherine the Great. In practical terms only if we have a welfare ration of 1.25 it means that unskilled worker could earn enough money in a construction season to provide subsistence. Wrapping up the discussion about working days, it's clear that this issue is common across all countries (Stephenson 2018, p. 129). Calculating annual income by multiplying daily earnings by a fixed number of working days is somewhat arbitrary. However, if we view this method as a convenient way to estimate actual daily earnings, it remains valid.

Figure 7 presents the attempt to integrate our results with those from the recent paper by Allen and Khaustova (2019). The graph displays our data spanning two centuries: up to 1815, we rely on our own dataset, while for the period after 1820, we incorporate prices and wages from Allen and Khaustova's dataset, released with their paper.

The graph illustrates that Allen and Khaustova's welfare ratio is significantly lower compared to our calculations after 1820. This difference arises because we substituted lamb for beef, as beef was more affordable and commonly consumed among peasants. Additionally, we excluded other expensive items like kerosene but maintained a calorie intake of 2,100. We adjusted our welfare calculations using our own subsistence basket, which includes essentials such as salt.

Figure 7. Welfare ratio of unskilled and skilled construction workers in the 18 and 19th centuries.



Sources: Appendix 3. Note: the dotted line indicates 1.25 welfare ratio based on real numbers of working days in Moscow in the 18th century in construction.

Each economic shock in the Moscow region temporarily improved the well-being of serfs, primarily due to labour shortages driving welfare increases. This pattern may reflect the unique characteristics of the construction industry. Ironically, the construction industry has driven wealth growth in Moscow for the past three centuries, including the 21st century. In this context, it is important to consider whether the construction industry is a reliable indicator of the welfare ratio in general.

Comparison of subsistence basket of Robert Allen to the realities of Moscow under serfdom.

There is an ongoing debate in economic history regarding the validity of Robert Allen's methodology (see Humphries and Schneider 2020, Humphries and Weisdorf 2019, and Stephenson 2018). I intend to assess how applicable Allen's proposed estimates are in the specific context of Russian serfdom. I acknowledge that my upcoming argument may raise doubts among those familiar with Russian history and surprise those who are not. Before I explain how a negative urban premium can occur, it is important to establish that the reality represented by the subsistence basket is not just a convenient method but one that accurately reflects the specific conditions in central Russia.

One significant factor influencing the results is the selection of goods for the basket. As shown in Figure 8, the original welfare ratio from Allen and Khaustova's paper is nearly half compared to the estimations in this paper. This trend is observed in other comparisons as well. For instance, wages in Polish cities appear much more favorable when using a bare-bone basket instead of a more comprehensive one that includes processed goods like bread and beer, which were cheaper in London (Malinowski 2016, p. 10–11).

Secondly, the conditions of the workers varied. British workers predominantly purchased their goods from the market. In contrast, Russian peasants produced a significant portion of what they consumed but still required money for taxes, quitrent to landowners, or purchases such as winter clothes and shoes.

Let us talk more about these considerations.

The choice of products for comparison significantly affects the outcomes of cross-country comparisons. It's important to note that Russia had cheaper products compared to other European countries. I will illustrate that a more accurate accounting of the expenses structure of a seasonal labourer

is unlikely to produce results that differ significantly from those obtained using the subsistence basket method.

One key point to discuss is that, as Van Zanden and Malinowski (2017) demonstrated through a comparison of urban wages and rural agricultural income, the income of agricultural workers should generally be viewed as household income, whereas urban workers' income is typically considered individual income. In the Russian context, however, both the income earned by a peasant who stays in farm and income from work outside agriculture should be treated as household income.

This is primarily because earnings from outside work were used by the family in the village to pay taxes and quitrent, and buy other necessities. Given the severe scarcity of cash in the Russian economy during the 18th and 19th centuries, any money earned outside agriculture was both expected and practically required to support the household. Additionally, Russian peasant families are well known for maintaining strong family connections and mutual support, so from both historical and cultural perspectives, the income of agricultural or urban workers should be viewed as household income.

A seasonal worker, whom I see as an extra person in the household as a son or young brother, leaving the farm needed to

a) support himself in the city in isolation from his own household, which provided him with cooked food and homemade clothes;

b) bring enough money to provide, combined with other family incomes (agriculture, handicrafts, transportation services), the needs of his household to pay a landowner and state.

I attempt to estimate the cost of living during the working season using historical data on nutritional standards. Naturally, these standards may diverge from actual practices, particularly regarding provisions in public institutions. However, they still serve as useful benchmarks for assessing the consumption patterns of seasonal workers.

The lowest provision levels were found in alms-houses. The standard allowance for a beggar in an alms-house was 1 kopeck per day, amounting to 3.65 rubles annually (Kozlova, 2010). Until 1744, this allowance remained unchanged due to financial constraints, not because of a decrease in the cost of living. A similar freezing of allowances was observed in the second half of the century.

Therefore, I consider these standards as indicative of consumption rates at the time. The annual allowance of 3.65 rubles closely approximates the cost of the subsistence basket calculated for 1703. However, it is higher, as the 3.65 rubles covered only food and clothing, while living in an alms-house also included costs for housing and firewood. In contrast, the subsistence basket costs 3.4 rubles, encompassing all living expenses, or 2.2 rubles excluding firewood.

In 1765, new regulations for the St. Petersburg and Moscow alms-houses of the College of Economy stipulated an annual allowance of 12 rubles or 3.3 kopecks per day for residents' provisions. Additionally, 3.6 rubles annually, roughly 1 kopeck daily, were allocated for clothes and bedding, such as a coat and a *caftan* made of grey cloth, four sets of shirts and pants, two pairs of wool stockings and two pairs of shoes. For heating, three logs each measuring 2.5 *sazhens* (1 sazhen = 2.13m) in length were provided per winter month, halved during the summer, costing 1.58 rubles per person (PSZ-RI-I, vol. 17, № 12334, pp. 59–63; vol. 44, part 2, pp. 44–45). Thus, the total expenditure per alms-house resident was 17.18 rubles annually, more than three times the cost of the subsistence basket in 1764. By 1786, the standards were slightly reduced, with residents receiving 3 kopecks daily for food and 1 kopeck for clothes, amounting to 14.6 rubles annually in Moscow alms-houses (RGADA, f. 1261, op. 12, d. 88, p. 13). In my calculations, the cost of the subsistence basket in 1786-1788 was roughly 13 rubles per year (see Appendix 1).

What could a person buy for 3 kopecks per day? In the off year of 1767, for 1.5 kopecks, a person could buy 1.5 rye loaves weighing a total of 2 pounds 45 spools (1.62 kg), and for an additional 1.5 kopecks, a pound of beef ('cheeks, rounds, loins') (Moskovskie vedomosti, 1767). By 1782, bread became more affordable, but beef prices had risen. For 1 kopeck, a person could buy rye bread weighing 2 pounds 55 spools (1.05 kg), but a pound of beef cost 2–2.5 kopecks (Gastev, 1841, p. 295; Zamechaniia k tablitsam, 1806, p. 169). The average annual price of a pound of rye bread from 1797 to 1800 fluctuated between 1.25 and 1.5 kopecks (Khlebopekarnyi 1899, p. 24). By the end of the century, 3 kopecks per day allowed a person to buy only 2 pounds of bread, which no longer sufficiently provided a minimum of 2,100 calories but could still stave off starvation.

Thus, money for feeding the elderly poor was allocated in plenty. This money allowed to consume nearly twice the minimum standard of 2,100 Cal prescribed in the basket, provided a person who purchased the cheapest goods like rye bread. Yet even the Synod Commission evidently expected that the elderly and sick would require more expensive and high-quality food.

In December 1807, a law was passed regulating the nutrition of children aged 2 to 5 years. Children of prisoners aged 2 to 5 years were to receive half the provisions of prisoners, which amounted to 1 *chetverik* of flour and $\frac{3}{4}$ *garntsa* quarters of groats per month. (PSZ RI-I, vol. 29, № 22730, p. 1363). The subsistence basket was always based on this assumption outlined by Robert Allen.

Next, let's consider provision standards for other unprivileged population, focusing on essential products according to these standards. Those who hired domestic workers on 'residential records' (*zhilye zapisi*) also believed that they should provide enough food to them. Comparing the wages of domestic workers 'under master's care' and the rest, E.I. Zaozerskaya found that food and clothing were estimated at 7 rubles per year in the first quarter of the 18th century (Zaozerskaia 1953, 449–450). Food in line with the abundant soldier's ration at that time cost 4.5 rubles. Then 2.5 rubles or more were spent on clothes.

But what exactly constituted the food ration? In 1775, A.S. Yartsov, head of the Olonets factories, argued for increased wages and estimated 'the most necessary provision' for an unmarried worker based on the soldier's ration. It consisted of approximately 21.6 poods of rye flour (353.8 kg), 1.5 poods of groats (24.57 kg), 13.5 poods (221.13 kg) of meat and fish, and 0.6 poods (9.83 kg) of salt, totaling 16.41 rubles (Strumilin, 1958, p. 392). In 1780, Prince Golytsyn allocated 18 rubles for food for an adult household serf, which is more than double the cost of the food portion of the subsistence basket in 1786 (Appendix 1, 4).

Thus, when those in power determined the nutrition standards for those who served them or the state, they assigned an abundant diet, which included a large amount of rye bread and a significant amount of meat.

What kind of diet did workers aim for when they had to buy food with their own money? Employees of the Moscow Cloth Yard provided a list of products in a complaint against the factory leaseholder who kept the prices of the 1740s in 1797: "[in the 1740s] a pood of rye flour cost 7 kopecks, a *chetverik* (26.2 liters) of buckwheat cost 10 kopecks, butter cost 3.5 kopecks, a pound of tallow candles cost 3 kopecks" (Tomsinskii, 1934, p. 190). In the 1750s, textile workers complained that they made cabbage and kvass in their yards, while those living at the factory 'could drink as much kvass as two kopecks worth during a hot day at work' (Zaozerskaia 1960, p. 363–364). They noted, 'for that much, a whole family at home could enjoy [*kvass*] for a week.'

For peasants, contemporaries reported a diet mainly consisting of bread, cabbage soup, occasional dairy products, and mushrooms throughout the 18th century. Meat was described as a rare holiday dish. However, A.T. Bolotov, a famous Russian writer on agriculture, indicated that the quantities were still larger than those in the bare-bone basket. In contrast, in the 1760s, observers often pointed out the insufficiency of the peasant diet (Durov 2002, 132–135; Bolotov 1766, p. 190, 203–204, 209, etc.).

All the products listed in the above sources are in good agreement with the later observations of A.N. Engelhardt, another famous Russian writer on agriculture: the food was mainly vegetable⁴. Fats, namely butter, vegetable oil, and lard, were predominantly used to increase the caloric value for hard work. Meat was not given much importance although it was, of course, appreciated as a delicious dish (Engelhardt 1999, pp. 231–236). The same ratio we found in the article by Chernevskii (Chernevskii 1941, p. 21 – 22). He described the diet for a peasant household of three adults and two kids which belongs to a middle class. They should have 10 *chetvert* of rye, 6 *chetveriks* of different grouts, 10 pounds of oil, 20 pounds of butter, 2 poods of salt, 1 *chetvert* of malt, and 3 fathom of firewood (*kubicheckie sazheni*). We should also be aware that different regions had varying norms. For example, in Tobolsk (Siberia), estimates by Link in the 1830s indicate that the average townsfolk consumed 163 pounds of meat, 4 barrels (*vedro*) of wine, and 4 fathoms (*sazheni*) of firewood. This is not surprising given the low cost of meat in Siberia (Link 1837, pp. 59–60).

Thus, we see that the diet did not fundamentally differ from the one laid down in the subsistence basket. Now we are assuming that a seasonal worker could try minimizing the costs of purchasing clothes

⁴ Potatoes joined bread and cereals only in the 19th century.

in the city by taking homespun clothes with him (Generalnoe..., 1784, p. 27). The second assumption is that the set of products from the subsistence basket was consumed not for a year, but for the working season from May 1 to October 24 (167 days). Thus, we get 1.66 kg of rye, 30 g of meat, and 30 g of oil (3,600 Cal) per day, quite enough for not just survival, but working. Increasing the cost of the food part of the subsistence basket by another 1/3 to account for spending on additional products (kvass, cabbage, vegetables) and unforeseen expenses, we get the following estimates of daily expenses: 2.5 kopecks per day in the beginning of the 1760s, 3.7 kopecks per day in 1777, 6.1 kopecks per day in 1786, from 9 to 11 kopecks per day in the years 1793 – 1800. For comparison, the rate of spending of relatively well-paid brickmakers of the Setunsky state factory in the 1770s was about 5.7 – 7.1 kopecks per day (Voronov 1957, p. 464-465).

The described level of spending refers only to the working season. Upon returning home, a peasant could consume his own bread, milk, and meat, use homemade needlework, and get firewood from the forest. However, at least in some places, bread had to be bought up. In addition, he had to repair buildings, maintain them in working order, and buy inventory as needed.

I estimate the welfare ratio in more detail for the 1780s. As shown in Figure 7, the advantages of moving to Moscow for work, indicated by a high welfare ratio, should be greater in the 1770s, 1810s, and 1870s. This is particularly true for the first two cases, which experienced the city's reconstruction after the fire. However, this increase was only temporary, and the welfare ratio eventually declined to a "normal" level of around 1. Therefore, the 1780s are a good example because they reflect a stable situation rather than a reaction to a shock.

Let us compare our calculations with expenses of an 'average' peasant household of a married couple with two children in the Tver' district (*uezd*) (1784). These calculations were done by contemporary officials. In total, the annual expenses of a household were 26.44 rubles (Generalnoe..., 1784, p. 25-26). Among those, 4.51 rubles were spent on local taxes and other household expenses, for the hut and other buildings for several years. Almost 1/3 (7.49 rubles⁵) was paid to the state and a landowner as poll tax and quitrent. 1.96 rubles were spent on a rig and a horse harness. Winter clothes and summer hats cost around 1 ruble. 1/6 (4.91 rubles) were spent on agricultural tools, various necessary iron homeware and wooden cutlery. It was supposed to spend 0.60 rubles on church. 3 rubles went 'to the wife and children', and the same amount could be saved or spent on 'unforeseen needs'. During a hard year, this amount, we suppose, could have been significantly reduced by refraining from allocating money to repair the yard, limiting expenses on harness, tools, and religious rites to the most necessary. It is conditionally accepted in calculations that it is possible to reduce 50% of such expenses. Then the costs would be around 12.25 rubles in total. So, the living costs of a family in the countryside could be estimated from 26 to 13 rubles per year in the 1780s.

This corroboration enables us to better understand the role of seasonal work earnings in the peasant household economy. We find that the actual amount of money used by a typical household in the 18th century aligns closely with estimates derived from Robert Allen's approach.

The Urban-rural wage premium in the context of Russian serfdom

The standard method for estimating the urban wage premium typically involves comparing the wages of city workers to those of agricultural workers in rural areas. However, due to insufficient data on agricultural wages and the subsistence economy⁶, I approached this issue by carefully estimating the incomes of both urban and agricultural workers, taking into account the specific realities of Russia and the nature of serfdom and household structures. The findings indicate a negative urban wage premium, revealing that the annual income from unskilled work in Moscow was lower than that from agriculture in the surrounding areas.

⁵ In this case the author of the description took quitrent norm 3 ruble per male. But in the same description he makes a remark that there existed another norm – 10 rules from 'venets' (married couple) (Ibid., 28). If we employ this norm poll tax and quitrent would require 11.4 rubles. We also should note that real payments were made according to part of *vyt'* which was a special taxation unit mostly connected with the acreage of land possessed by a household (L. S. Prokof'eva, *Krest'ianskaia obshchina v Rossii vo vtoroi polovine XVIII – pervoi polovine XIX veka* (L., 1981), 54 - 93). In our calculations we assume that for average household paid a quitrent according to norm for three males or one married couple.

⁶ It refers to an economic system where goods are produced primarily for personal consumption rather than for trade or market exchange.

How much could a serf or a peasant earn from agriculture in non-black soil regions? Considering the amount of arable land listed as no less than a tithe (*desiatina*) in one field per male serf in the ‘*cameral description*’, the depicted family must have had an allotment of at least 2 tithes in each field. Given a crop yield as 3 of rye and as 4 of oats and a seeding rate of 10 – 13 chetveriks of rye, 4 quarters of oats per tithe, the resulting net harvest (excluding grain put off for seeds) is 40 – 52 chetveriks (poods) of rye, 16 quarters (approximately 84.8 poods) of oats (Generalnoe..., 1784, p. 23). According to the average annual Tver prices for 1783⁷, this harvest could cost 32.27 rubles. From the perspective of nutrition, the harvest was enough to provide a family at a rate of 24 poods per adult and 12 poods for a child – a total of 72 poods. In addition, farming provided hay, straw, and feed for livestock, and allowed for non-grain crops to be grown along. Thus, in normal conditions, for a family of Tver peasants with one adult male worker, farming was advantageous, followed by transportation services in wintertime, which, according to the author of the description, could provide 15-20 rubles per winter. Paired with the selling of a small surplus of oats, butter, and livestock (no more than for 6 rubles), as well as home canvas (80 arshin for 4 or 5 rubles), the household could make ends meet, albeit by a very small margin.

Even during a bad year with a yield with 2 of all crops, the net harvest was supposed to deliver 20 – 26 chetveriks of rye, 8 quarters of oats – a total of 61 – 67 poods of grain, which could support a family of four (at the end of the 19th century 20 poods of bread per year was considered quite an acceptable rate). Given successful winter enterprise and canvas weaving, the possibility to earn more than 20 rubles and make ends meet, while somewhat reducing expenses, retained.

Let’s compare these estimates with the wages of qualified bricklayers in the early 1780s. He could earn up to 40 rubles (earning 50 rubles and spending 10 rubles on food and clothes per season). In times of low prices, a peasant engaged in seasonal work who brought home 40 rubles could feed his family and still have around 27 rubles left, which would be just enough to cover the family’s expenses. Additionally, his wife could earn about 5 rubles from weaving, which could suffice to buy forage for a horse. Although we should remember that bricklayer wages were high at the beginning of the 1780s due to the Moscow authorities’ demand to transform Moscow into a European city. Thus, even in a very favourable year, seasonal work brought in about the same amount as agricultural work for a nuclear family. Moreover, the peasant did not need to travel or spend time acquiring the skills required to become a qualified bricklayer. So, the gains from agriculture were comparable or even higher in case of nuclear family.

However, given the presence of one additional worker in the family (the unmarried brother of the breadwinner or the eldest son), his participation in the summer work was evidently advisable. He could earn 9-12 rubles employed as a horse breeder (*konevod*) per summer. Meanwhile, he could take part in mowing and reaping while on breaks (Generalnoe..., 1784, p. 25-26). Leaving for Moscow as a labourer for the season from May 1 to October 24, getting paid 15 kopecks a day, he could earn 22.5 rubles in 150 days, deducting the cost of living in the capital from 3 to 7 kopecks a day, the net profit would amount to 12-18 rubles. Considering the expenses for the trip and the lost opportunity to take part in haymaking and reaping, the gain from labouring in Moscow was either absent or insignificant. Working at a brick factory as an unskilled labourer from May 10 to August 15 would have brought a little more than 19 rubles, deducting 7 rubles for accommodation – a little more than 12 rubles, however, prolonging work as a labourer until the 20th of October (63 working days excluding Sundays and holidays), would make it possible to increase this amount by another 9.45 rub. (5 – 7.5 rubles, deducting expenses for accommodation). In total, we consider about 17 – 19.5 rubles, which the seasonal worker could bring home. In that case, a trip to Moscow appears to be more profitable than more traditional horse breeding.

Thus, for a worker from an average Tver peasant household, work in Moscow could be worthwhile if he performed at least partially qualified work and was not the only one in the household. This conclusion is in line with Bolotov’s idea that working out of native village (*otkhod*) is an option only for peasants with big families (Bolotov, 1766, p. 198, Prokhorov, 1997, 331). However, in the event of the ruin of the peasant household, labouring (including in Moscow) could become a necessary means of survival. Example of such situation described in biography of prominent Petersburg merchant (and former peasant of Iaroslavl’ district) Aleksander Berezin. After the death of his father Berezin flew to

⁷ 1.75 ruble per chetvert’ of rye, 0,98 rub. per chetvert’ of oats (Mironov, *Khlebnye tseny*, 227–228)

Petersburg and started to work as a house servant (Berezin 1879, p. 299). In this case, taxes were redistributed within the community, and expenditure items were reduced.

Let's estimate what contribution to the household budget a peasant from the Sheremetyevs' estate *Molodoy Tud* of the Rzhev county could make if he went to Moscow and was hired there as an unskilled labourer. The choice of the estate is convenient in several ways. First, for the first half of the 1780s, the evaluation of the monetary part of the peasant households in the Tver county can be the same as in Rzhev county. Second, the quitrent for 1762 – 1800 is known for this estate. Third, it was indeed a region with a developed seasonal work practice, although Moscow was not among the places where Rzhev peasants usually looked for work.

The size of the monetary part of the household budget in my calculations consists of

1. the quitrent and the poll tax per household;
2. the expenses on harness and 'all sorts of iron things'⁸;
3. the personal expenses of a labourer during the working season. Due to the lack of data, expenses for a hat and mittens, a priest, women and children, and "unforeseen needs" were excluded. These expenses are partially filled in by the price of 3 quarters of rye.

Table 4. The comparison of calculated income of seasonal unskilled labourer and cash expenses of a household of five people (Rzhev)

Years	Cash expenses of a household and expenses on accommodation in Moscow (rubles)	calculated income of a seasonal unskilled labourer for 150 working days (rubles)	Income/expenses
1762 – 1766	9.13	24.00	2.63
1772 – 1776	13.38	34.50	2.58
1777 – 1781	14.08	27.00	1.92
1782 – 1785	21.42	24.00	1.12
1789 – 1793	27.94	22.50	0.81
1794 - 1797	36.47	33.00	0.90

Sources: Appendix 5.

Overall, we obtain the same trend as from comparing the earnings of a day labourer with a tripled subsistence basket. A particularly strong decline was in the 1790s. As we can see, the use of a more plausible number of working days and the inclusion of expenses specific to Russian peasants (rent, iron products) did not significantly influence the assessment results of the dynamics of the level of wages applying to day labourers in Moscow.

Conclusion

The methodology proposed by Robert Allen allows for the estimation of the welfare ratio in 18th-century Russia. Despite the scattered nature of the data, new findings can prompt significant adjustments. The welfare ratio in Russia during the third quarter of the 18th century was comparable to that of large Polish cities and slightly higher than in many other European cities. By the beginning of the 19th century, the welfare ratio in Russia was significantly higher than in the rest of Europe. The 1770s and 1810s were particularly favourable for both skilled and unskilled workers. However, the welfare ratio decreased slightly in the last quarter of the century, which does not align with trends in carpenters' wages calculated by B.N. Mironov or weavers' wages estimated by A.V. Kovalchuk. Our research suggests that skilled labor, including carpenters, grew much faster than unskilled labor during this period.

Comparing the welfare ratio of daily laborers with the budget of a peasant household in non-black soil region, like Tver, reveals that the monetary part of the budget and the subsistence basket changed similarly during the second half of the 18th century. Thus, using the subsistence basket as a

⁸ Unfortunately, we have no sufficient data on harness or homeware prices. So, we deflated sum for homeware in 1783 budget in accordance with prices on price scissors from Boris Mironov database (Boris Mironov, Peter Lindert, Russia. Prices and wages 1590 – 1871, *Global Price and Income History Group*: <http://gpih.ucdavis.edu/Datafilelist.htm>).

deflator does not significantly distort the assessment of the purchasing power of seasonal workers' earnings.

This study highlights the importance of considering life cycles and regional contexts in research on living standards in non-industrialized societies. The latest GDP per capita series by Broadberry and Korchmina show stagnation from the late 18th century to the 1870s, with no significant impact from events like the Napoleonic Wars. In contrast, Moscow's welfare ratio for laborers was notably affected by these wars, underscoring the significance of regional perspectives, which overall GDP data for Russia often overlooks.

Speculating on the motivations of serfs and peasants in pre-industrial times is challenging, mainly due to the lack of qualitative sources. However, my research on welfare ratios, combined with findings from Russian historians that up to 40% of serfs and peasants worked in cities, offers a new perspective. As noted, serfs were more likely to migrate to the city by choice rather than through systematic pressure from landlords. This study tentatively suggests that serfs and peasants made economically rational choices based on market incentives: they migrated to Moscow when wages rose but preferred to stay on their estates when wages fell. This rational approach allowed them to maintain their subsistence through agriculture, with city work primarily feasible for additional household members, such as unmarried sons.

The research indicates that in the latter half of the 18th century, living standards in Russia were adequate to support a family, despite the country being economically poorer than much of Europe. There was a notable, albeit temporary, spike in the welfare ratio during the 1800s and 1810s. The graph also includes data from Jamaica, representing a society based on slavery, where the welfare ratio—averaged between slaves and impoverished whites for a single year—shows that the ratio for slaves was only 0.4. This comparison underscores that serfs, despite facing their own challenges, were in a significantly better position than slaves in terms of living standards.

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Average Annual Nominal Prices of Products in Moscow, 1706 – 1875.

1738	0.80	0.01	0.04	0.24	2.40	1.29	0.03	1.34	0.52	3.47	0.89	0.19	4.01	0.83
1741				0.24		1.70	0.03	1.20						
1744	1.30													
1745	1.00													
1746	0.90						0.04							
1747	0.90													
1748	1.00													
1749	1.70													
1750	1.20													
1751	1.15													
1752	1.10													
1753	0.90													
1757	0.75						0.04	1.80		4.03				
1758	0.90													
1759	0.85													
1760	0.76													
1761	0.74					1.60								
1762	1.03	0.01	0.05	0.40	3.07	1.82	0.03	2.20	0.67	1.00	0.26	0.20	4.20	0.87
1763	1.05	0.01	0.05	0.40	3.15	2.04	0.02	2.80	0.69	1.08	0.28	0.21	4.32	0.90
1764	1.03	0.01	0.05	0.40	3.10	1.68	0.03	1.60	0.60	1.33	0.34	0.20	4.25	0.76
1765	1.22						0.05	1.60	0.44					
1766	2.28						0.03							
1767	2.59	0.01	0.06	0.50	6.54	1.60	0.05	1.60	0.73	2.00	0.52	0.39	8.18	1.47
1768	2.30						0.05							
1769	2.10		0.05	0.40		2.65	0.06			1.05				
1770	1.54		0.05	0.40		2.47				1.17				
1771	1.34													
1772	1.73		0.03	0.40		2.27				3.35	0.86			
1773	1.79													
1774	2.03													
1775	2.65		0.06	0.40		2.50	0.09			2.40				
1776	2.18					2.00				4.32				
1777	1.66	0.03	0.06	0.40	4.71	2.00	0.05	1.20	0.73	1.50	0.39	0.29	6.12	1.09
1778	1.43		0.02	0.40				4.00		1.65				
1779	1.86						0.06							
1780	1.98	0.05				2.80	0.05			1.98				
1781	1.52						0.04							
1782	3.13	0.06	0.12							1.50				
1783	3.13	0.02	0.12											
1784	3.13	0.02	0.12											
1785	2.84	0.04	0.12											
1786	3.11	0.02	0.07	0.40	7.79	3.69	0.04	1.60	1.01	1.29	0.33	0.50	10.60	1.87
1787	5.16	0.01	0.09	0.40	12.30	4.00	0.03	1.60	1.00	4.11	1.06	0.87	18.28	3.19
1788	3.50	0.02	0.08	0.40	8.88	4.38	0.03	1.60	1.00	2.10	0.54	0.60	12.57	2.10
1789	3.50	0.03	0.09	0.40	8.94	4.75	0.08	1.60		1.53	0.40	0.52	11.00	1.82
1790	3.25	0.05	0.11	0.40	8.93	4.70				1.00	0.26			
1791	2.66	0.05	0.11	0.40	7.77	4.85				1.10	0.28			
1792	3.30	0.05	0.12	0.40	9.15	5.30	0.06	3.64		1.00	0.26	0.47	9.88	1.41
1793	4.24	0.06	0.13	0.40	11.39	5.01	0.05	3.64	1.44	3.43	0.88	0.69	14.40	1.92
1794	4.00	0.04	0.15	0.40	10.94	4.47	0.04	3.64	1.28	1.20	0.31	0.63	13.16	1.68

1795	4.36	0.04	0.16	0.40	11.87	9.67	0.04	4.30	2.16	4.40	1.13	0.76	15.92	1.96
1796	5.52	0.04	0.17	0.40	14.43	5.66	0.08	4.84	1.81	2.27	0.58	0.84	17.67	2.24
1797	4.71	0.04	0.14	0.40	12.33	4.80	0.10	4.87	1.85	1.20	0.31	0.72	15.21	2.17
1798	4.05	0.06	0.14	0.40	11.19	4.80	0.08	4.87	1.68	1.20	0.31	0.66	13.84	1.82
1799	3.99	0.05	0.14	0.44	10.89	6.00	0.09	4.90	1.94	2.43	0.63	0.67	14.13	1.68
1800	4.54	0.05	0.19	0.40	12.69	5.74	0.09	5.10	1.91	6.09	1.92	0.83	17.36	2.04
1801	5.18	0.04	0.17	0.40	13.65	5.46	0.09	5.00	1.86	6.02	1.73	0.86	18.11	2.16
1802	4.20	0.04	0.13	0.41	11.09	6.05	0.09	4.62	1.96	5.94	1.53	0.73	15.31	2.00
1803	5.95	0.04	0.16	0.40	15.16	6.64				1.20	0.31			
1804	4.00													
1805	3.50	0.04	0.13	0.40	9.67	6.20	0.10	6.40	2.20	9.62	2.48	0.72	15.07	2.09
1806	4.75	0.02	0.23	0.40	13.21	6.10	0.14	5.05	2.35	3.68	0.95	0.83	17.34	2.33
1807	5.37	0.04	0.16	0.40	13.89	6.00	0.14	3.70	2.23		1.19	0.87	18.18	2.20
1808	6.00	0.06	0.09	0.40	14.57	6.40				5.58	1.44			
1809	6.50	0.06	0.25	0.40	17.67	7.60	0.25	1.00	3.05	11.13	2.87	1.18	24.76	2.00
1810	7.80	0.07	0.36	0.40	21.81	12.00	0.19	9.00	3.92	11.00	2.84	1.43	30.00	1.78
1811	8.67			0.40				4.80		12.00	3.09			
1812	11.67	0.10	0.44	0.40	31.31	9.47	0.26			11.32	2.92			
1813	12.40		0.43			11.56	0.32			16.00	4.12			
1814						19.39	0.26			17.77	4.58			
1815	9.20	0.09	0.45	1.72	26.69	18.05	0.32	19.80	6.69	16.42	4.23	1.88	39.49	1.42
1816														
1817														
1818														
1819														
1824	10.29	0.10				12.16		10.24		20.00	5.15			
1825	10.29	0.15	0.36			10.24		10.24						
1826	10.29	0.14	0.35			10.56		9.76						
1827	10.29	0.14	0.34			10.72	6.28	9.28	46.64	22.00	5.67			
1828	7.35	0.09	0.29	2.30	21.11	11.20	6.90	9.28	51.13	18.50	4.77	3.85	80.86	3.93
1829	8.82	0.11	0.31	2.30	24.71	10.72	7.53	9.28	55.48	16.50	4.25	4.22	88.66	4.31
1829	6.25	0.08	0.29	2.00	18.55	10.75	7.53	9.28	55.48	14.50	3.74	3.89	81.66	3.97
1830	10.29	0.10	0.33	2.30	27.93	10.72		10.08						
1831	13.23	0.12	0.41	2.30	35.32	11.20		10.72		18.00	4.64			
1832	17.64	0.13	0.42			13.44		10.88						
1833	23.52	0.15	0.44			15.68	7.53	10.88	56.39	19.00	4.90			
1834	20.58	0.15	0.43	2.40	51.50	13.12		10.56						
1835	16.17	0.15	0.42			10.72		10.08						
1836	11.76	0.16	0.41	2.40	32.78	12.00	6.59	9.76	49.09	19.00	4.90	4.34	91.11	4.49
1837	11.76	0.17	0.44	2.30	33.16	12.00		9.76						
1838	10.29	0.18	0.46	2.20	30.48	11.52		9.76						
1839	10.29	0.12	0.31			7.68		8.16						
1840	8.82	0.06	0.15			4.00		6.56						
1841	10.29	0.06	0.16			3.84		4.96						
1842	5.88	0.06	0.15			3.84		3.36						
1843	7.35	0.05	0.14			3.84		3.20						
1844	8.82	0.04	0.15			3.84		3.04						
1845	5.88	0.04	0.15			3.84		2.88						
1846	2.94	0.04	0.14	0.72	8.73	3.84	1.57	2.72	11.88					
1847	4.41	0.05	0.14			3.84	0.00	2.72						

1848	2.94	0.05	0.13			3.68	0.00	2.72						
1849	2.94	0.05	0.12			3.68	0.00	2.88						
1850	2.94	0.05	0.10	2.20	8.97	3.52	0.00	2.88						
1851	4.41	0.05	0.16			3.36	1.88	2.88	14.02	11.50	2.96			
1852	4.41	0.04	0.15	0.70	11.98	3.36	0.00	2.88						
1853	4.41	0.05	0.14	0.70	11.94	3.52	0.39	2.72	3.53					
1854	4.41	0.05	0.13	0.40	11.67	3.68	0.42	2.56	3.74	8.00	2.06	0.87	18.35	3.14
1855	4.41	0.04	0.14	0.40	11.67	3.36	0.44	2.72	3.84	8.00	2.06	0.88	18.45	3.16
1856	5.88	0.04	0.15	0.60	15.00	4.16	0.42	2.88	3.85					
1857	5.88	0.05	0.15	0.60	15.17	4.96	0.42	3.68	4.04					
1858	5.88	0.06	0.17	0.60	15.48	4.96	0.34	3.84	3.50					
1859	5.88	0.07	0.18	0.75	15.81	4.96	0.34	3.68	3.49	8.50	2.19	1.07	22.56	3.86
1860	7.35	0.08	0.25	0.75	19.78	5.44	0.34	3.84	3.57					
1861	5.88	0.09	0.22	0.75	16.47	6.24	0.34	3.84	3.70					
1862	5.88	0.08	0.23	0.75	16.45	5.92	0.34	3.84	3.65					
1863	4.41	0.07	0.24	0.75	13.39	5.60	0.30	3.84	3.33					
1864	4.41	0.09	0.23	0.80	13.45	4.32	0.30	3.20	3.07					
1865	4.41	0.07	0.23	0.80	13.30	4.64	0.31	3.84	3.24	9.00	2.32	0.94	19.80	2.96
1866	7.35	0.09	0.22	0.80	19.66	5.28	0.31	3.84	3.34	10.00	2.58	1.28	26.86	4.01
1867	7.35	0.12	0.22	0.80	20.00	5.44	0.31	3.84	3.37	8.50	2.19	1.28	26.84	4.01
1868	7.35	0.12	0.22	1.20	20.18	5.44	0.31	4.16	3.39	11.00	2.84	1.32	27.73	4.14
1869	7.35	0.12	0.23	0.80	20.02	5.60	0.31	4.48	3.44	11.50	2.96	1.32	27.75	4.15
1870	7.35	0.11	0.23	0.80	20.03	5.60	0.31	4.80	3.47	10.30	2.65	1.31	27.46	3.95
1871	8.82	0.10	0.24	0.80	23.09	5.60	0.31	5.12	3.50	7.50	1.93	1.43	29.94	4.31
1872	7.35	0.16	0.25	0.80	20.79	5.76	0.32	4.64	3.53	8.00	2.06	1.32	27.71	3.99
1873	0.00	0.14	0.31	0.80	5.77	5.76	0.34	5.12	3.73	7.00	1.80	0.57	11.87	1.71
1874	8.82	0.15	0.33	0.80	24.74	5.76	0.34	5.12	3.73	6.00	1.55	1.50	31.51	4.54
1875	8.82	0.14	0.30	0.73	24.29	5.76	0.34	4.64	3.69	7.00	1.80	1.49	31.27	4.50

Sources: RGADA, f. 16, op. 1, d. 582, l. 176–228; f. 237. op. 1 d. 181; f. 280. op. 6, d. 3172, 3356; f. 1204, op. 1, d. 199, 19303, 19305, 19311, 19312, 19314, 19315, 19317, 19319, 19324, 19279, 19281, 19284, 19286, 19297, 19298, 19299, 19301, 19377; f. 1263. op. 1. d. 6440; f. 280. op. 5. d. 108; f. 292. op. 1, d. 4, 11, 13, 15; f. 297. op. 1, d. 1067; f. 297. op. 1, d. 1068; Moskovskie vedomosti, 1786, 1801, 1805. Zanechania k tablitsamv Moskve s 1782 po 1827, TsGA, f. 421, op. 1, d. 5312., TsGA, f. 421, op. 1, d. 5312., <https://lib-fond.ru/lib-cgam/421/f-421-op-1-537/>. Prices from the 1820s are taken from the realesaed data (Allen Khaustova). Yellow cells are highlited because the prices are calculated. For all data see the online appendix as an excel file.

Appendix 2

Wages of different workers in Moscow.

year	Black smith	Black smith	Black smith	Brick layer	Brick layer	Brick layer	Brick maker	Brick maker	Brick maker	Carp enter	Carp enter	Carp enter	skilled labour	skilled labour	Labou rer	unskilled labour	unskilled labour
	obs	daily wage	daily wage	obs	daily wage	daily wage	obs	daily wage	daily wage	obs	daily wage	daily wage	daily wage	daily wage	obs	daily wage	daily wage
		roubles	in silver		roubles	in silver		roubles	in silver		roubles	in silver	roubles	in silver		roubles	in silver
1701															1	0.06	0.0148
1704				1	0.02	0.0049							0.02	0.0049			
1718	1	0.02	0.0039	1	0.03	0.0074							0.02	0.0057			
1722															2	0.03	0.0061
1724															2	0.05	0.0111

1726				1	0.10	0.0246							0.10	0.0246			
1729															1	0.02	0.0039
1730															1	0.08	0.0198
1731															1	0.01	0.0030
1734															1	0.02	0.0041
1736															2	0.01	0.0012
1737									1	0.01	0.0036		0.01	0.0036	2	0.01	0.0034
1738															2	0.04	0.0095
1739															1	0.02	0.0041
1740															1	0.01	0.0032
1741				1	0.12	0.0297							0.12	0.0297			
1749	2	0.36	0.0893										0.36	0.0893			
1756															1	0.02	0.0049
1757				1	0.15	0.0371							0.15	0.0371			
1759				2	0.24	0.0594							0.24	0.0594			
1762															1	0.04	0.0099
1764							1	0.57					0.57	0.1029	6	0.11	0.0204
1765																	
1766									1	0.25	0.0450		0.25	0.0450			
1767				1	0.16	0.0288							0.16	0.0288	1	0.05	0.0090
1769									1	0.07	0.0125		0.07	0.0125	1	0.05	0.0089
1772															1	0.08	0.0141
1773				1	0.27	0.0480							0.27	0.0480			
1776															1	0.20	0.0356
1777				1	0.45	0.0807			1	0.03	0.0053		0.24	0.0430	3	0.18	0.0315
1778				1	0.45	0.0807							0.45	0.0807	16	0.16	0.0282
1779				4	0.37	0.0658			6	0.29	0.0514		0.33	0.0586	21	0.13	0.0232
1780				1	0.45	0.0798			3	0.18	0.0327		0.32	0.0562	9	0.14	0.0258
1781				1	0.45	0.0798							0.45	0.0798			
1784															6	0.14	0.0249
1785															7	0.14	0.0239
1786															2	0.10	0.0175
1787															3	0.18	0.0306
1788															3	0.14	0.0233
1789				1	0.25	0.0391							0.25	0.0391	2	0.17	0.0258
1790									6	0.48	0.0707		0.48	0.0707	8	0.17	0.0249
1791				1	0.41	0.0579							0.41	0.0579			
1793															3	0.11	0.0143
1794				3	0.16	0.0195			2	0.15	0.0185		0.15	0.0190	2	0.25	0.0308
1795				3	0.19	0.0235							0.10	0.0127			
1796				1	0.30	0.0429			1	0.80	0.1143		0.55	0.0786	1	0.25	0.0357
1797				4	0.23	0.0302							0.23	0.0302			
1798				3	0.15	0.0179							0.15	0.0179	1	0.10	0.0119
1800				1	0.61	0.0724							0.61	0.0724	8	0.23	0.0271
1801																	
1802																	
1803																	
1804																	
1805																	
1806									1	0.55	0.0664		0.55	0.0664		0.40	0.0483

1807																	
1808																	
1809				1	1.35	0.0802							1.35	0.0802			
1810							1	1.35		1	1.40	0.0665	1.38	0.0653			
1811																	
1812																	
1813																	
1814																	
1815									1	2.50	0.0900		2.50	0.0900		1.86	0.0670
1819																	
1820																	
1821																	
1822																	
1826											1.75	0.0851	1.75	0.0851		0.75	0.0365
1827											0.70	0.0340	0.70	0.0340		0.60	0.0292
1828																	0.0000
1829																	0.0000
1830																	0.0000
1831																	0.0000
1832											1.80	0.0907	1.80	0.0907		1.00	0.0504
1833																	0.0000
1834											1.90	0.0958	1.90	0.0958		0.97	0.0489
1835																	0.0000
1836																	0.0000
1837											1.90	0.0958	1.90	0.0958		0.90	0.0454
1838																	0.0000
1839											1.80	0.0907	1.80	0.0907		1.00	0.0504
1840											0.50	0.0882	0.50	0.0882		0.30	0.0529
1841											0.60	0.1058	0.60	0.1058		0.30	0.0529
1842											0.50	0.0882	0.50	0.0882		0.30	0.0529
1843											0.40	0.0706	0.40	0.0706		0.25	0.0441
1844											0.50	0.0882	0.50	0.0882		0.33	0.0582
1845											0.65	0.1147	0.65	0.1147		0.30	0.0529
1846											0.45	0.0794	0.45	0.0794		0.25	0.0441
1847											0.65	0.1147	0.65	0.1147		0.35	0.0617
1848											0.50	0.0882	0.50	0.0882		0.25	0.0441
1849											0.45	0.0794	0.45	0.0794		0.25	0.0441
1850											0.40	0.0684	0.40	0.0684		0.25	0.0428
1851											0.40	0.0684	0.40	0.0684		0.25	0.0428
1852											0.50	0.0855	0.50	0.0855		0.25	0.0428
1853											0.45	0.0770	0.45	0.0770		0.25	0.0428
1854											0.45	0.0770	0.45	0.0770		0.25	0.0428
1855											0.50	0.0855	0.50	0.0855		0.30	0.0513
1856											0.60	0.1026	0.60	0.1026		0.30	0.0513
1857																	
1858											0.80	0.1368	0.80	0.1368		0.40	0.0684
1859											0.85	0.1454	0.85	0.1454		0.50	0.0855
1860											0.90	0.1345	0.90	0.1345		0.50	0.0747
1861											1.00	0.1494	1.00	0.1494		0.60	0.0896
1862																	

1863											1.00	0.1494	1.00	0.1494		0.70	0.1046
1864											1.00	0.1494	1.00	0.1494		0.75	0.1121
1865											1.10	0.1643	1.10	0.1643		0.75	0.1121
1866											1.10	0.1643	1.10	0.1643		0.70	0.1046
1867											1.10	0.1643	1.10	0.1643		0.70	0.1046
1868																	
1869																	
1870																	
1871											1.17	0.1685	1.17			0.87	0.1254
1872											1.20	0.1734	1.20			0.81	0.1161
1873											1.16	0.1674	1.16			0.75	0.1086
1874											1.15	0.1656	1.15			0.86	0.1245
1875											1.24	0.1788	1.24			0.94	

Source: RGADA, f. 237, op. 1, d. 181; f. 16, op. 1, d. 482; f. 280, op. 5, d. 108; op. 6, d. 3172, 3356; f. 292, op. 1, kn. 4, 11, 13, 15; f. 297, op. 1, d. 1067, 1068; f. 1191, op. 1, d. 216, l. 23, 24, 48; f. 1204, op. 1, d. 199, 19303, 19305, 19311, 19312, 19314, 19315, 19317, 19319, 19324, 19279, 19281, 19284, 19286, 19297, 19298, 19299, 19301, 19377; f. 1261, op. 12., d. 88, f. 1263, op. 1, d. 6440; 6224, PSZ I, n. 13256, Voronov, 1957. After 1825 the data is taken from Allen and Khaustova (2019).

Appendix 3

Welfare Ratio of Moscow Unskilled and Semiskilled Labourers

Year	the cost of annual subsistence basket per person (rubles)	unskilled labor (daily wages in rubles)	skilled labour (daily wages in rubles)	the cost of annual subsistence basket per family (rubles)	Welfare ratio for unskilled labour	Welfare ratio for skilled labour	how many days need unskilled to work to earn subsistence	how many days need skilled to work to earn subsistence	Welfare ratio for unskilled labour (Allen and Khaustova, 2019)	Welfare ratio for carpenters (Allen and Khaustova, 2019)
1718	4.77		0.02	14.32		0.40				
1722		0.03								
1724		0.05								
1726			0.10					123.63		
1728	4.12			12.36	0.62	2.02	353.23			
1729		0.02								
1730		0.08								
1731		0.01								
1734		0.02								
1736		0.01								
1737		0.01								
1738	4.01	0.04		12.02	0.80		313.58			
1740		0.01								
1741			0.12					100.17		
1749			0.36							
1756		0.02								
1757			0.15							
1759			0.24							
1762	4.20	0.04		12.60	0.79		315.11	52.52		
1763	4.32			12.97						
1764	4.25	0.11		12.75	2.22					
1765			0.61							
1766			0.25							

1767	8.18	0.05	0.16	24.54	0.51	1.63	490.73	153.35		
1769		0.05	0.07							
1772		0.08								
1773			0.27							
1776		0.20								
1777	6.12	0.18	0.24	18.35	2.41	3.29	103.89	75.99		
1778		0.16	0.45							
1779		0.13	0.33							
1780		0.14	0.32							
1781			0.45							
1784		0.14								
1785		0.14								
1786	10.60	0.10		31.81	0.79		318.05			
1787	18.28	0.18		54.83	0.84		299.07			
1788	12.57	0.14		37.72	0.93		267.81			
1789	11.00	0.17	0.25	33.00	1.25	1.89		150.87		
1790		0.17	0.48							
1791			0.41							
1792	9.88									
1793	14.40	0.11		43.20	0.65		386.88			
1794	13.16	0.25		39.47	1.58	0.00	157.87			
1795	15.92		0.17	47.76		0.90				
1796	17.67	0.25	0.55	53.00	1.18	2.59	212.00	96.36		
1797	15.21		0.23	45.64		1.26				
1798	13.84	0.10	0.15	41.51	0.60	0.88	415.11	284.32		
1799	14.13			42.40						
1800	17.36	0.35	0.61	52.08	1.68	2.92	148.80	85.71		
1801	18.11			54.32						
1802	15.31			45.94						
1803										
1804										
1805	15.07	0.30		45.20	1.66		150.67			
1806	17.34	0.40	0.55	52.02	1.92	2.64	130.05	94.59		
1807	18.18			54.54						
1808										
1809	24.76		1.35	74.28		4.54		55.02		
1810	30.00		1.38	89.99		3.82		65.45		
1811										
1812										
1813										
1814										
1815	39.49	1.86	2.50	118.48	3.92	5.28	63.70	47.39		
1816										
1817										
1818										
1819										
1824										
1825										
1826		0.75	1.75							
1827		0.60	0.70							

1828	80.86									
1829	81.66									
1830										
1831										
1832		1.00	1.80							
1833										
1834		0.97	1.90							
1835										
1836	91.11									
1837		0.90	1.90							
1838										
1839		1.00	1.80							
1840		0.30	0.50							
1841		0.30	0.60							
1842		0.30	0.50							
1843		0.25	0.40							
1844		0.33	0.50							
1845		0.30	0.65							
1846		0.25	0.45							
1847		0.35	0.65							
1848		0.25	0.50							
1849		0.25	0.45							
1850		0.25	0.40							
1851		0.25	0.40							
1852		0.25	0.50							
1853		0.25	0.45							
1854	18.35	0.25	0.45	55.05	1.14	2.04	220.18	122.32	0.40	0.60
1855	18.45	0.30	0.50	55.36	1.35	2.26	184.52	110.71	0.45	0.55
1856		0.30	0.60							
1857										
1858		0.40	0.80							
1859	22.56	0.50	0.85	67.68	1.85	3.14	135.35	79.62	0.60	1.30
1860		0.50	0.90							
1861		0.60	1.00							
1862										
1863		0.70	1.00							
1864		0.75	1.00							
1865	19.80	0.75	1.10	59.41	3.16	4.63	79.22	54.01	1.00	1.50
1866	26.86	0.70	1.10	80.59	2.17	3.41	115.12	73.26	1.00	1.50
1867	26.84	0.70	1.10	80.51	2.17	3.42	115.01	73.19	1.00	1.50
1868	27.73									
1869	27.75									
1870	27.46									
1871	29.94	0.87	1.17	89.83	2.42	3.26	103.15	76.78	0.90	1.50
1872	27.71	0.81	1.20	83.12	2.43	3.62	103.09	69.03	1.10	1.60
1873	11.87	0.75	1.16	35.61	5.30	8.16	47.21	30.63	1.20	1.70
1874	31.51	0.86	1.15	94.54	2.29	3.04	109.34	82.21	1.10	1.50

Appendix 4

Payments to Golitsyn's Household Serfs in 1780

	Age	People in household	Salary	For clothes	For food	Total	Occupation
Mavra Chenysheva	60	1	16.00	10.00	24.00	50.00	Nurse
Andrei Smirnov	24	1	30.00	26.00	24.00	80.00	under-tutor
Mitrofan Rodionov	28	1	12.00		18.00	30.00	Valet
Ivan Gusarev	19	1	12.00		18.00		Hairdresser
Fedor Beliaev	20	1	10.00		18.00		Hairdresser
Mikhailo Bashmachnikov	21	1	10.00		18.00	28.00	teacher's assistant
Afanasii Ukrainian boy	16	1	7.00		18.00	25.00	a boy
Ivan Kiselev	14	1	7.00		18.00	25.00	a boy
Dmitrii Evstifiev	48	3	8.00		36.00	44.00	watchmen of the chancellery and their wives who work as laundresses
Wife of Dmitrii Evstifiev	42						
Fedor, son of Dmitrii Evstifiev	13				9.00	9.00	
Leontii Stepanov	37	2	8.00		36.00	44.00	
Wife of Leontii Stepanov	29						
Pension for former nurse Anna		1	50.00			50.00	
Chief of the chancellery Semen Smirnov	45	5	300.00		120.00	420.00	Chief of the chancellery from simskie household serfs
Semen's wife	42						
Three sons of Semen							
Nikita Golubev	30	1	16.00		18.00	34.00	a scribe
Avram Podkatov	20	1	12.00		18.00	30.00	a scribe
Semen Zolotarev	28	1	12.00		18.00	30.00	on errand
Mikhail Balashev	27	1	12.00		18.00	30.00	on errand
Larion Mizintsev	31	1	6.00	6.00	18.00	30.00	a horseman
Arefei Bychkov	34	1	6.00	6.00	18.00	30.00	a horseman
Stepan Sokolov	38	3	8.00		36.00	44.00	watchmen of the chancellery and their wives who work as laundresses
Wife of Stepan Sokolov	29						
Ivan, son of Stepan Sokolov	7				9.00	9.00	
Gavrilo, son of Stepan Sokolov	3	1					
Egor Sadovoi	30	1	7.00		18.00	25.00	
Afanas'ev Klim	25	1	7.00		18.00	25.00	
Iakov Lebedev	25	1	7.00		18.00	25.00	a watchman and a gardener's pupil

Nikolai Rychagov	24	1	7.00		18.00	25.00	a cook's pupil
Evdokim Pil'shchikov	54	2	7.00		23.00	30.00	janitor for the backyard
Wife of Evdokim Pil'shchikov	56						
Karp Zelentsov, widowed	50	1	7.00		18.00	25.00	janitor for the backyard
Fedor Samoilov	62	2	4.00	4.00	7.00	15.00	old-aged [retired]
Wife of Samoil Fedorov	40						
Fedot Usahev	58	1	3.00	3.00	4.00	10.00	
bailiff Andrei Beliaev and his wife	26	2	50.00			50.00	bailiff
Il'ia Alpat'ev with his wife and daughter	62	3	24.00			24.00	Was liberated from serfdom but did not manage to find work because of old age; without occupation
Aleksei Ivanovich Menshchikov	63	1	12.00			12.00	he is blind and does not see at all
Ananii Ermolaev, his wife and two children	43	4	15.00			15.00	without occupation; was a footman
Anna Ivanovs, wodow with two children	39	3	18.00			18.00	her husband was a forester
Aleksei Nevkin, vice ensign and his wife,	59	2	12.00			12.00	lives in a stone house
Andrei Ivanov with his wife and daughter	59	3	12.00			12.00	Former horseman; caretaker
Fedorov Ivan with his wife and six children	56	8	15.00			15.00	carpenter; caretaker
Ivan Nikitin with his wife	64	2	13.00			13.00	old horseman
Fedor Gusarov with his wife and daughter	65	3	13.00			13.00	old horseman
Vasilii Morozov	62	1	10.00			10.00	old horseman
Il'ia Trofimov Orlov	64	1	6.00			6.00	old horseman
Fedot Tsarev with his wife	47	2	8.00			8.00	Received a passport to work outside manor but failed to find work and returned to ask for a pay
Mikhail Koz'min with two grandsons	61	4	10.00			10.00	old horseman1
Vasily Makarov with his wife	59	2	10.00			10.00	old horseman
Semen Akulov with his wife and five children	52	7	22.00			22.00	former coachman, church caretaker; his children are learning / training

Aleksei Seliverstov with his wife	60	2	10.00			10.00	Was a scribe and now has no occupation
Avdot'ia Nikitichna	59	1	6.00			6.00	eldery widow
Avdot'ia Il'ina	68	1	6.00			6.00	eldery widow
Marfa Lavrent'eva	79	1	6.00			6.00	eldery widow
Katerina Ivanova	59	1	6.00			6.00	eldery widow
Avdot'ia Borodulina	70	1	6.00			6.00	eldery widow
Avdot'ia Dmitrieva Sharnikova	41	5	6.00			6.00	eldery widow
four children of Avdot'ia Sharnikova			12.00			12.00	literate
widow of Kareshennikov	36	4	10.00			10.00	
three children of widow. The eldest.	9						literate
Stepan Nikitich Proba with his wife and two children	59	4	15.00			15.00	shop guard
a widow and her son	49	2	10.00			10.00	a her husband was a horseman
Ivan Ivanov and his wife	31	2	24.00			24.00	gardener
Maksim Petrov and his son	36	3	15.00			15.00	gardener with an eye decease
Kalina Kuzmin	22	1	10.00			10.00	gardener's pupil
8 people learning to read and to write	от 17 до 9	8	56.00			56.00	
Gavrilo Il'in	58	1	8.00			8.00	teacher
10 elmsmen	72 - 32	10	55.00			55.00	almsmen
5 women living in alms-house	75 - 46	5	27.50			27.50	almsmen
Sergei Fedorov with his wife and daughter	30	3	100.00			100.00	bailiff
Iakov Naumov	20	1	22.00			22.00	scribe
Boris Bumashkin with his wife	50	2	20.75			20.75	not specified
widow	50	1	7.00			7.00	
widow	51	1	7.00			7.00	
widow	39	1	7.00			7.00	
Total		148				1,924.25	

Source: RGADA, f. 1263, op. 1, d. 6365. Calculations are ours.

Appendix 5

Relation of seasonal worker (*otkhodnik*) Income to Money Required for Peasant Household of Five Persons

Year	Quitrent per married couple	Poll tax (3 male serf)	Costs of <i>otkhodnik</i> 's accommodation outside his household	Cost of 100 <i>Dvoetes</i> Nails	Cost of Iron Home-work	Cost of 3 <i>chetvert'</i> of rye	Total	Average daily wage	Estimated income for 150 days	Income / Needs Ratio
	ruble	ruble	ruble	ruble	ruble	ruble	ruble	ruble	ruble	
1762 – 1766	2.18	2.10	4.14	0.095	1.41	3.96	13.79	0.11	16.5	1.20
1772 – 1776	2.60	2.10	7.30	0.14	2.08	6.18	20.26	0.16	24.0	1.18
1777 – 1781	3.30	2.10	7.30	0.14	2.08	5.07	19.85	0.14	21.0	1.06
1782 – 1785	4.80	2.10	7.00	0.33	4.91	4.56	23.37	0.14	21.0	0.90
1789 – 1793	6.00	2.10	14.59	0.40	5.95	11.43	40.07	0.17	25.5	0.64
1794 – 1797	8.65	3.00	20.61	0.35	5.21	13.95	51.42	0.17	25.5	0.50

Sources: Prokof'eva, *Krest'ianskaia obshchina v Rossii vo vtoroi polovine XVIII – pervoi polovine XIX veka*, 112–115; *General'noe soobrazhenie po Tverskoi gubernii...*, 122; appendix 1; Boris Mironov, Peter Lindert, Russia. Prices and wages 1590 – 1871, *Global Price and Income History Group*: <http://gpih.ucdavis.edu/Datafilelist.htm> (Accessed on 15 January 2022); appendix 3.